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Text Guided Cross Attentive Multimodal Learning with Visual Feature Modulation for Automated Skin Lesion Detection

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Abstract

Automated skin lesion detection is essential for early dermatological diagnosis. Most deep learning algorithms employ dermoscopic images and ignore the clinical context of dermatologists. This constraint decreases the robustness and interpretability, particularly in visually ambiguous instances. An explainable multimodal cross attention framework that merges clinical text with dermoscopic images to increase diagnostic accuracy and semantic grounding in automated skin lesion detection. A Text-Guided Cross-Attentive Visual Feature Network multimodal learning architecture (TG-CAVNet) using Bio-ClinicalBERT based clinical text encoding and EfficientNet-B4 visual feature extraction is proposed. The framework uses text-guided channel-wise feature modulation, text-queried cross-attention for semantic-spatial alignment and adaptive multi-stream fusion to merge complementary representations. The model was trained end-to-end using hybrid focal and cross-entropy losses. On a multimodal dermoscopic dataset of 6,194 aligned image-text samples, TG-CAVNet outperforms state-of-the-art multimodal baselines with 90.75% accuracy and a macro Jaccard score of 0.82. Ablation investigations are performed that confirmed the separate and synergistic impacts of the components, whereas attention visualizations improved interpretability. Text-guided cross-attentive multimodal learning improves the performance and explainability of automated skin lesion identification. The robust and clinically interpretable decision-support framework TG-CAVNet demonstrates the necessity of integrating semantic clinical context with visual analysis in dermatological AI systems.

Keywords: Dermoscopic Imaging, Multimodal Fusion, Clinical Text Integration, Visual Encoding, Medical Imaging

1. Introduction

The rising global incidence of melanoma and non-melanoma skin malignancies and the fact that early diagnosis improves treatment results and survival rates make automated skin lesion detection an important part of dermatological decision support. Dermoscopy based lesion assessment is the clinical norm, although physician competence and subjective interpretation determine its accuracy. Thus, computer-aided diagnostic systems using deep learning for skin lesion analysis have been extensively studied for their objectivity, reproducibility, and scalability [1]. Recent research shows that convolutional neural networks (CNNs) can learn discriminative visual patterns from dermoscopic images to execute dermatologist-level tasks in controlled settings, suggesting their therapeutic use [2].

Despite these developments, vision-only deep learning algorithms have significant limitations in dermatological settings. Different illumination, imaging instruments, lesion anatomy, and artefacts such as hair, rulers, and colour calibration markers cause intra- and inter-class variability in dermoscopic pictures [3]. Vision-only models are susceptible to these characteristics and often lack contextual cues beyond pixel-level appearance, reducing their robustness and generalizability across datasets and clinical contexts [4]. Dermatologists rarely use image appearance alone for diagnosis; therefore, visual approaches do not reflect their process.

Dermatological diagnosis involves visual inspection and textual clinical information regarding lesion colour progression, border characteristics, anatomical location, patient demographics, and pertinent medical history. These textual descriptors provide a high-level semantic context for the visual interpretation of distinguishing visually similar lesions. Journal studies have demonstrated that adding clinical metadata or textual information to photographs improves diagnostic accuracy and model stability [5]. Many multimodal techniques use simple feature concatenation or late fusion strategies, which fail to simulate fine-grained text-visual interactions and limit interpretability [6].

Owing to these issues, current research has focused on text-guided visual reasoning, where textual input conditions modulate or query visual aspects during representation learning. Attention-based multimodal models can link semantic text cues with spatially relevant visual regions, improving the performance and explainability of medical image analysis [7]. However, the integration of clinical text into dermatological vision models, particularly through dynamic visual encoding mechanisms, has not been studied [8].

To overcome these limitations, Text-Guided Cross-Attentive Visual Feature Network (TG-CAVNet), a unique multimodal system for automated skin lesion diagnosis is proposed in this research that directly incorporates the effect of clinical text on visual feature learning. TG-CAVNet uses a pretrained biomedical language encoder to extract semantic embeddings from clinical text and a deep convolutional backbone to derive multiscale visual representations from dermoscopic images. Two complementary mechanisms use textual embeddings: a text-guided gating feature modulation module, which adaptively recalibrates channel-wise visual features based on clinical semantics and a text-queried cross-attention module, which enables text-driven focus on diagnostically relevant image regions. The gated and attended visual features were merged using multi-stream fusion and sent to a classification head for lesion prediction.

The key contributions of this research are:

- A text-guided visual feature modulation technique that uses channel-wise attention and a gating mechanism to adaptively recalibrate visual feature representations using clinical textual semantics during visual encoding that improves visual ambiguity.
- A text-queried cross-attention module allows clinical text embeddings to query spatial visual characteristics for fine-grained semantic alignment between textual descriptors and diagnostically relevant image regions.
- A multi-stream visual feature fusion framework that adaptively fuses gated and cross-attended visual characteristics to harness complementing channel-level and spatial-level information.
- Normalized weighted fusion stabilizes heterogeneous visual stream integration, reduces feature scale imbalances, and retains discriminative representations during multimodal learning.
- Attention heatmap visualization is used to automatically detect skin lesions and provide clinically relevant text-guided visual reasoning insights.

2. Related Works

Automated skin lesion detection is a clinically relevant issue because the early and accurate detection of malignant lesions affects patient outcomes and treatment methods. Melanoma and other skin cancers are major public health issues, and diagnostic variability during visual assessment leads to misclassification and delayed intervention. This prompted the development of computer-aided diagnostic systems to support dermatological decision-making and reduce observer-dependence [9]. Deep learning based algorithms have superseded handmade feature methods in dermoscopic image analysis because they can build hierarchical visual representations directly from data and perform better across vast and diverse datasets.

2.1 Skin Lesion Detection Automation

CNN-Based Dermoscopy Analysis: Owing to their ability to simulate complicated visual patterns, convolutional neural networks (CNNs) dominate automatic lesion categorization and segmentation. Models like ResNet, DenseNet, and EfficientNet [10] can capture discriminative texture, colour, and shape information from dermoscopic images better than classical image processing [11]. These vision-only CNN models are susceptible to imaging circumstances, lesion heterogeneity, and acquisition abnormalities, which limit their practical applicability. Enhancements such as significant data augmentation and ensemble learning are image-centric and do not leverage the clinical semantic context, which is crucial for dermatological diagnosis [12].

Multi-Scale Attention-Based Vision Models: Multi-scale representation learning captures lesion characteristics at multiple spatial resolutions to overcome the limitations of CNNs. Lesion border delineation and classification have improved with feature pyramid networks and deep supervision algorithms that incorporate global contextual information with fine-grained local details [13]. In parallel, attention-based vision models emphasize conspicuous lesion locations and informative feature channels using spatial and channel-wise attention [14]. These attention mechanisms improve visual feature quality but are driven exclusively by visual statistics and are uninformed by clinical semantics, restricting their ability to adapt feature extraction to the diagnostic context. This issue is overcome by text-guided channel-wise feature modulation in TG-CAVNet, which allows clinical descriptions to directly influence the visual feature calibration.

2.2 Multimodal Learning in Medical Imaging

Multimodal learning uses heterogeneous input sources to create rich and robust representations. When visual evidence is confusing or insufficient, combining images with clinical metadata, laboratory findings, and written reports improves the accuracy of medical imaging diagnosis [15]. These findings emphasize the relevance of non-visual information in automated diagnosis.

Image-Text Fusion Methods: Early, late and hybrid image - text fusion methods are used in medical imaging. Early fusion methods integrate input or shallow feature modalities for joint representation learning; however, modality imbalance occurs when high-dimensional visual data are combined with sparse textual embeddings [16]. Late fusion techniques aggregate modality-specific predictions at the decision level, simplifying training but severely limiting cross-modal interactions and semantic alignment [17]. Hybrid fusion techniques integrate modalities at intermediate network stages for flexibility and performance [18]. However, most hybrid techniques use static fusion mechanisms that do not actively change the impact of textual information on visual representation learning. Text-queried cross-attention and adaptive multi-stream fusion allow clinical text to actively direct spatial attention and feature integration during inference in TG-CAVNet.

2.3 Modulating Attention and Features:

Modern deep learning architectures use attention and feature modulation for selective focusing and contextual adaptation. Squeeze-and-excitation [19] and feature-wise linear modulation (FiLM) [20] recalibrate feature maps by assigning adaptive weights to channels based on contextual relevance, improving the discrimination and robustness of medical image analysis. Context-aware visual encoding is possible with feature modulation and auxiliary inputs, although most applications use global image statistics or metadata instead of rich clinical texts.

Multimodal Alignment Cross-Attention: By permitting one modality to attend to another, cross-attention mechanisms represent diverse modalities [21]. Cross-attention aligns textual or metadata representations with spatial visual information in multimodal medical imaging, enabling text-driven localization of diagnostically relevant picture regions and attention heat maps for interpretability [22]. Despite these gains, previous studies have examined cross-attention and channel-wise modulation separately. Dermatological image analysis has not yet included text-guided channel-wise feature modulation, text-queried cross-attentive spatial reasoning, and adaptive multi-stream fusion with normalization. TG-CAVNet unites these methods into a multimodal framework for automated skin lesion identification to address this gap.

Table 1. Summary of Literature

S. No.	Methodology	Modalities Used	Dataset	Key Contributions	Limitations
1	Multi-resolution EfficientNet ensemble + metadata fusion [23]	Image + Metadata	ISIC	Strong CNN ensemble, Improved classification via multi-resolution features	No clinical text modelling, No cross-attention, Early fusion only, No interpretability
2	Attention-guided multimodal fusion using EfficientNet + Dempster–Shafer rule [24]	Image + Metadata	PAD-UFES, HAM10000	Uncertainty-aware fusion, Attention for metadata	No free-text No channel-wise modulation No semantic-guided visual reasoning
3	FDUM-Net (FPN + U-Net) for segmentation [25]	Image	ISIC	Strong segmentation using enhanced U-Net structure	Segmentation only No multimodal fusion No text guidance or semantics

4	SCSONet with spatial–channel synergy [26]	Image	ISIC	Improved spatial-channel optimization for segmentation	Unimodal No text No cross-attentive fusion No diagnostic classification
5	Early multimodal CNN fusion (image + metadata) [6]	Image + Metadata	ISIC	Demonstrated early benefit of multimodality	Outdated CNNs No transformers No semantic alignment No text-guided reasoning
6	Large vision–language foundation model [8]	Image + Free-text + Metadata	Large multi-institutional datasets	Large-scale pretraining, Strong multimodal reasoning	Extremely large model Hard to deploy No explicit text-guided feature modulation targeted to dermatology
7	MiSC hybrid multimodal system (image + text) [27]	Image + Clinical text	Private dataset	Shows power of text–image fusion	Uses late fusion Lacks cross-attention No channel recalibration from text
8	Context-aware hierarchical fusion for X-ray reading [28]	Image	X-ray datasets	Contextual integration improves classification	No multimodal cues No NLP/clinical text No text-guided spatial attention
9	Cross-attention-based fusion for imaging + tabular metadata [29]	Image + Metadata	4D CTP	Demonstrates impact of cross-attention	No free-text No feature modulation Not dermatology-specific
10	**Proposed TG-CAVNet:** Text-guided cross-attentive multimodal learning with visual feature modulation	Image + Clinical Free-Text	6,194 multimodal dermoscopic samples	Bio-ClinicalBERT encoding Text-guided channel-wise gating Text-queried cross-attention Multi-stream fusion Explainable heatmaps	Dermatology model combining text-guided gating + cross-attention

As shown in Table 1, multimodal dermatology frameworks either fuse dermoscopic images with structured metadata or focus solely on visual segmentation without semantic clinical cues. This results in models that lack deep semantic alignment between clinical knowledge and visual patterns. Cross-attention integrates imaging with auxiliary information in modern multimodal clinical systems. However, they employ tabular metadata rather than vivid clinical writing. Few studies use free-text clinical descriptions, but these approaches use late fusion, do not recalibrate channel-wise features, and do not inject text-derived semantic queries into spatial visual attention maps. Table 2 shows that earlier works use CNN-based visual learning, and some use attention techniques, but none incorporate free-text semantics, feature modulation, and cross-attentive visual querying. TG-CAVNet directly covers this gap using Bio-ClinicalBERT based text encoding, text-guided channel-wise recalibration, text-queried cross-attention for semantic–spatial alignment, and adaptive multi-stream fusion. This unified approach improves multimodal synergy, diagnostic accuracy, and clinically meaningful interpretability for automated skin lesion identification.

Table 2. Comparison of existing methods against TG-CAVNet architectural components

Ref.	CNN-Based Visual Learning	Attention Mechanism	Text / Clinical Semantics	Feature Modulation	Cross-Attention
Gessert et al., 2020 [23]	□	□	□	□	□
Aboulmira et al., 2025[24]	□	□	□	□	□
Sharen et al., 2024 [25]	□	□	□	□	□
Chen et al., 2024 [26]	□	□	□	□	□
Yap et al., 2018 [6]	□	□	□	□	□
Yan et al., 2025 [8]	□	□	□	□	□
Mohamed et al., 2025 [27]	□	□	□	□	□

Chang et al., 2024 [28]	□	□	□	□	□
Amador et al., 2025 [29]	□	□	□	□	□
Proposed TG-CAVNet	□	□	□	□	□

3. Materials and Methods

3.1. Overall System Architecture

The proposed multimodal framework Text-Guided Cross-Attentive Visual Feature Network (TG-CAVNet) integrates dermoscopic images and clinical textual descriptions to automatically detect skin lesions. The proposed multimodal learning technique increases the representation quality by combining visual and written information. Dermoscopy images show lesions in detail, but textual descriptors provide contextual semantic information that images cannot. This approach emphasizes diagnostically significant visual structures by dynamically aligning various modalities using cross-attention. Multimodal interaction improves feature discrimination and classification ability, as demonstrated in the experiments.

The core learning problem is defined as follows. Given a matched dermoscopic image and clinical text description, the model must learn to identify the correct lesion class using information from both modalities. The TG-CAVNet model diagnoses as a conditional prediction over the combined image–text input space, where the text actively shapes visual feature extraction and interpretation. The model was trained end-to-end by minimizing a hybrid loss that penalized inaccurate predictions and prioritized difficult and minority class samples, with weight decay regularization preventing overfitting. The text encoder, visual encoder, gating module, cross-attention module, fusion module, and classification head were optimized together so that the text and visual pathways co-adapt to accurately classify the lesions. This ensured that the learned representations reflected the complementary diagnostic information carried by both modalities.

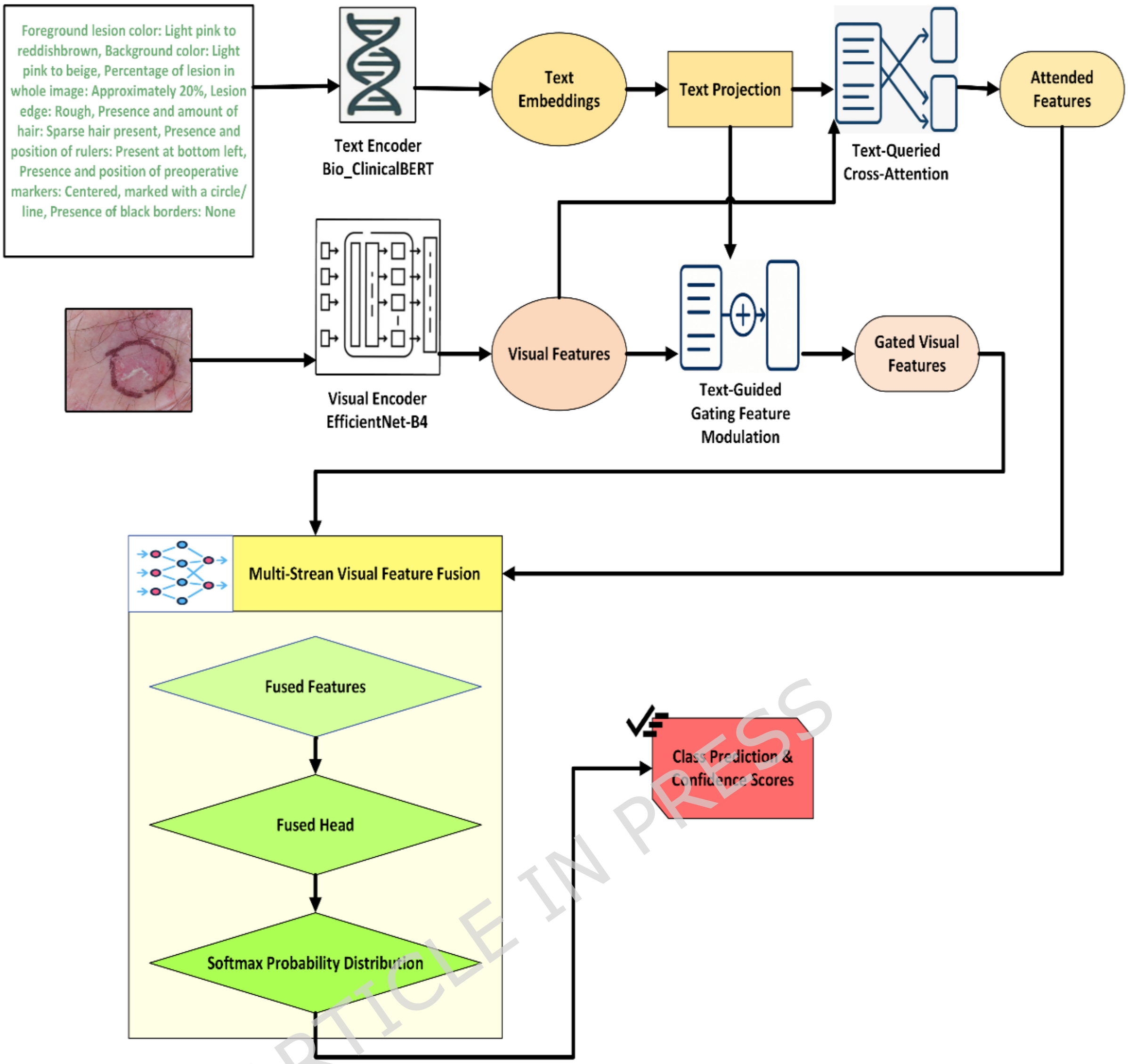


Figure 1. Overall architecture of the proposed TG-CAVNet Multimodal Framework

Figure 1 shows the architecture of TG-CAVNet that uses text-guided feature modulation, cross-attention and adaptive multi-stream fusion to fuse textual and visual representations after processing them separately to extract modality-specific representations. The textual approach begins with clinical language detailing the lesion colour, size, border irregularity and artefacts. A pretrained Bio-ClinicalBERT text encoder generates contextualized text embeddings with rich biomedical semantics from this input data. A text projection layer aligns the dimensions of these embeddings with the visual feature space, enabling downstream modules to interact with visual representations. Dermoscopy images were processed using the visual pathway, utilizing an EfficientNet-B4 backbone to extract high level visual properties, such as lesion texture, colour distribution, and structural patterns. Visual feature maps form the basis of modulation and attention-based reasoning.

Two text-guided strategies improve visual representation learning in the TG-CAVNet. First, the text-guided gating feature modulation module generates adaptive gating signals to calibrate the visual features using projected text embeddings. Clinically significant feature channels are accentuated, whereas noisy or less informative channels are muted through channel-wise modulation. This allows the visual encoder to dynamically alter its representations based on the clinical semantic context, incorporating diagnostic priors into visual features. Second, the text-queried cross-attention module enables fine-grained text-image semantic alignment. This module queries visual feature maps as keys and values, using projected text embeddings. Cross-attention computes attention weights to emphasize the spatial parts of the image that are most relevant to clinical descriptions, creating attended visual features that capture text-driven spatial significance. This technique identifies textually influenced diagnostic picture regions to improve interpretability.

The architecture incorporates gated and attended visual features using a multi-stream visual feature fusion module. TG-CAVNet balances channel-wise calibrated and spatially attended features using adaptive fusion rather than concatenation. This fusion phase creates a robust visual representation of the global semantic conditioning and local discriminative patterns. Finally, the fused representation is sent to a classification head with fully connected layers and a softmax function to generate class predictions and confidence scores representing benign or malignant. The end-to-end design of TG-CAVNet ensures that textual information influences visual encoding, attention and fusion, enabling context-aware, interpretable and robust automatic skin lesion diagnosis. A hybrid loss function that addresses class imbalance and optimizes stability was used to train the model end-to-end. The total training loss if given by,

$$L_{total} = \lambda_{focal} \cdot L_{focal} + \lambda_{ce} . L_{ce} \tag{1}$$

where the focus loss L_{focal} down weights easy samples and the cross-entropy L_{ce} term maintains classification stability. During optimization, the three loss components complement and stabilize each other. Focal loss reduces class imbalance by down-weighting easily correctly classified samples and focusing gradient updates on challenging and minority-class malignancy instances, pushing the model to improve clinically relevant samples. By preventing early training epoch parameter adjustments from hard instances that dominate the gradient signal, cross-entropy stabilizes the classification. Label smoothing with $\varepsilon = 0.1$ softens the target distribution, preventing overconfident logit values and maintaining the calibrated output probability estimates. This prevents the saturation of the Softmax function, which would otherwise reduce the gradient magnitude to zero in the later epochs. The smoothly decreasing training and validation loss curves in Figure 10 show that the focal loss sharpens the focus on hard samples, the cross-entropy maintains the gradient breadth across all samples, and the label smoothing prevents logit explosion. These three mechanisms operate simultaneously on different failure modes, resulting in a more stable convergence than any single loss term.

Weighting coefficients λ_{focal} and λ_{ce} control the component contributions during training. Figure 2 shows the internal design and information flow of the proposed TG-CAVNet, which corresponds to the high-level workflow in Figure 1. The model is a sequential, modular pipeline that blends clinical text and dermoscopic images through well-built components. After independent multimodal feature extraction to preserve modality-specific properties, text-guided processes condition visual representation learning on clinical semantics. Clinical literature influences global and local visual encoding through channel-wise feature modulation and cross-attentive spatial reasoning. Before lesion categorization, adaptive multi-stream fusion integrates these complementary representations. The following subsections provides the detailed modelling of each architectural component in the TG-CAVNet framework.

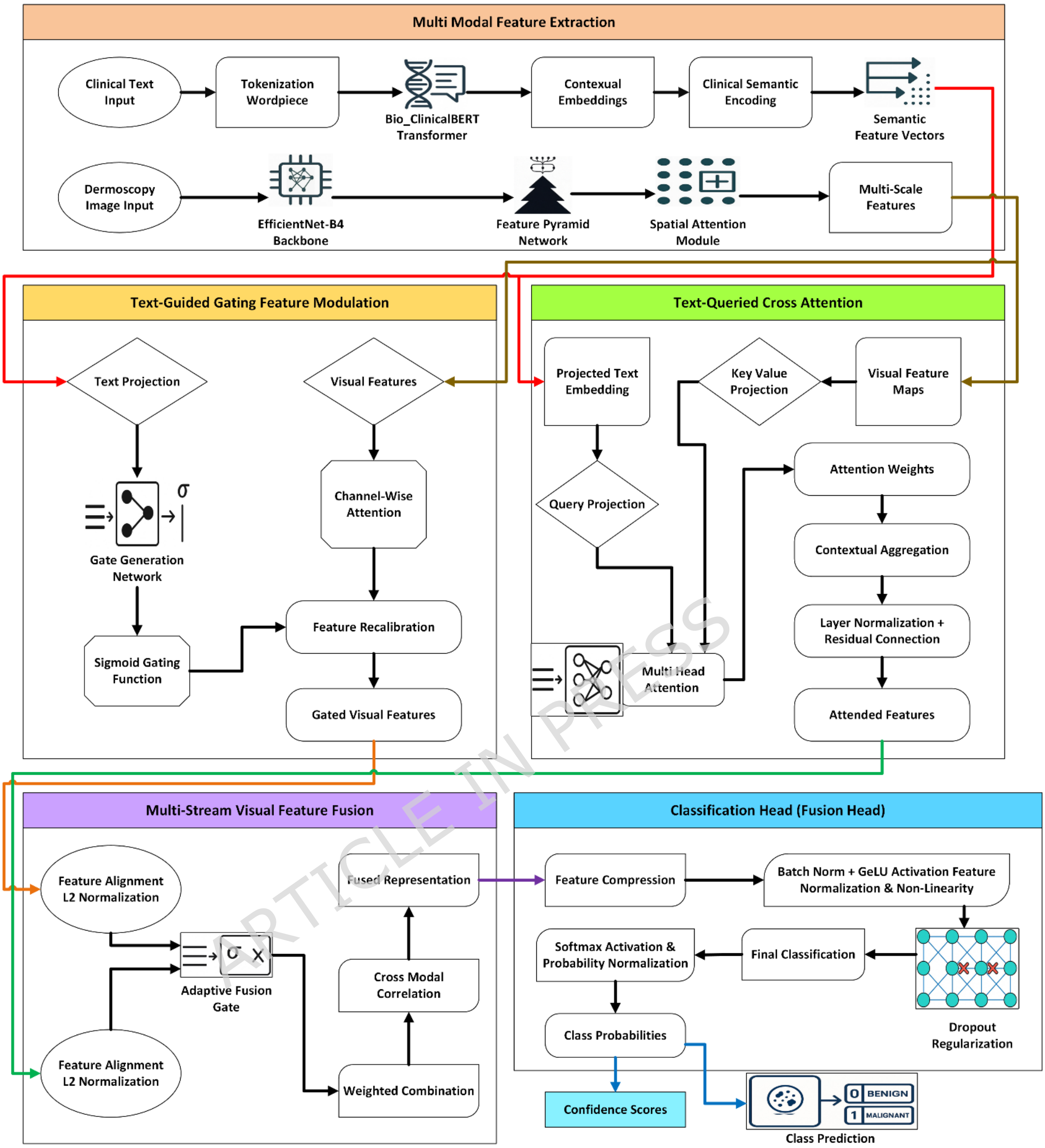


Figure 2. Workflow of the proposed TG-CAVNet multimodal Framework

Dermoscopy visual features and contextual clinical text descriptors were integrated into the TG-CAVNet multimodal representation learning architecture. Let I_i be the dermoscopic image and T_i the clinical textual description of the i^{th} skin lesion. Two modality-specific encoders were used to extract latent representations from these inputs. The visual encoder f_v (.) creates a visual feature representation F_{v_i} from the dermoscopic image, whereas the textual encoder f_t(.) uses Bio-ClinicalBERT to convert clinical language into a contextual semantic representation F_{t_i} . F_{v_i} stores spatial lesion characteristics, including colour changes, border abnormalities, and structural patterns, whereas F_{t_i} encodes dermatological narrative-derived semantic clinical descriptors. Formally, modality-specific feature extraction is represented as $F_{v_i} = f_v(I_i)$ and $F_{t_i} = f_t(T_i)$.

The multimodal learning framework aims to model the interplay between these two modalities in a single representation. Text-guided gating feature modulation, text-queried cross-attention, and multi-stream visual feature fusion enable this interaction in TG-CAVNet. These modules gradually correlate semantic textual information with visual feature representations, allowing the model to dynamically recalibrate visual characteristics based on clinically relevant language patterns. The multimodal representation F_{fused} includes visual lesion features and surrounding semantic information. The classification head uses this fused representation to accurately forecast skin lesions, allowing the TG-CAVNet to employ multimodal information for better diagnosis.

The architecture is robust for both input modalities. The EfficientNet-B4 backbone, pretrained on diverse large-scale data, produces stable feature maps that do not change drastically with minor pixel-level changes for visual perturbations, such as lighting variation, imaging artifacts, and augmentation noise. The spatial attention module suppresses non-lesion background noise before the cross-modal interaction. Bio-ClinicalBERT's bidirectional contextual encoding ensures that the global [CLS] representation captures sentence-level diagnostic meaning rather than relying on any single word, making it resilient to local textual changes, such as synonym substitution or minor phrasing variation. Importantly, the adaptive fusion weight dynamically adjusts the contribution of each modality stream during inference; therefore, if one modality has a higher uncertainty or noise for a sample, the network can naturally upweight the more reliable stream, providing a soft self-correcting mechanism for image and text perturbation scenarios.

3.2. Multi-Modal Feature Extraction

In the proposed TG-CAVNet framework, dual-stream multimodal feature extraction encodes clinical textual descriptions and dermoscopic images separately before enabling regulated cross-modal interactions. This solution preserves modality-specific properties while providing clinical text semantic assistance to impact visual thinking. The retrieved semantic and visual representations underpin text-guided feature modulation, cross-attentive alignment and adaptive multi-stream fusion.

3.2.1. Clinical Text Processing with Bio-ClinicalBERT

For dermatological diagnosis, clinical language descriptions provide high-level semantic clues, including lesion colour, border irregularity, asymmetry, anatomical location, and clinical progression. Clinical text is

explicitly incorporated as a first-class modality in the proposed TG-CAVNet architecture to provide contextual semantic representations for learning visual features. Each clinical text description T describes the lesion-specific features.
When dealing with specialized medical language and variable-length clinical narratives, wordpiece tokenization is used to break text into sub words while maintaining semantic meaning. Text is a sequence of such tokens as given in Eq. (2), where N is the token count of the sequence and w_i is the i^{th} wordpiece token. This tokenization technique effectively manages unusual or compound dermatological words.

$$T = \{w_1, w_2, ...w_N\} \quad (2)$$

Bio-ClinicalBERT, a transformer-based language model pretrained on huge biomedical literature and clinical notes, encodes the tokenized sequence. Bio-ClinicalBERT captures token bidirectional contextual relationships using stacked transformer encoder layers and multi-head self-attention.

Algorithm TG_CAVNet_FORWARD(*I, T*)
Input: Dermoscopic image $I \in R^{(384\times384\times3)}$,
Clinical text description *T*
Output: Class prediction ŷ ∈ {Benign, Malignant},
Probability distribution *p*,
Novel intermediate outputs $\{\gamma, \alpha, F_{fused}\}$

// ============ PHASE 1: FEATURE EXTRACTION ============
1: T_{tokens} ← WordPieceTokenize(*T*) // Tokenization
2: E_t ← Bio_ClinicalBERT(T_{tokens}) // Transformer encoding
3: $t \leftarrow E_t$ [0] // [CLS] token pooling
4: $t_s \leftarrow W_t \cdot t + b_t$ // Semantic projection // $t_s \in R^{512}$
5: I_{norm} ← Normalize(*I*) // ImageNet normalization
6: F_v ← EfficientNet_B4(I_{norm}) // Base features
7: $F_v^{(l)}$← FPN(F_v) // Multi-scale pyramid
8: A_s ← SpatialAttention(F_v) // Spatial attention
9: $F_v^{sa} \leftarrow A_s \odot F_v$ // Spatially attended features

// ============ PHASE 2: NOVEL COMPONENTS ============
10: F_v^{gate}, γ ← CONCEPT_TO_GATE(t_s, F_v^{sa})
11: $t_p \leftarrow W_p t_s + b_p$
12: F_v^{att} ← TEXT_QUERIED_CROSS_ATTENTION(t_p, F_v^{sa})
13: F_{fused} , α ← ADAPTIVE_MULTI_STREAM_FUSION(F_v^{gate}, F_v^{att})
// ============ PHASE 3: CLASSIFICATION ============
14: z_1 ← GeLU(BN($W_{c1} \cdot F_{fused} + b_{c1}$)) // First compression

15: $z_1 \leftarrow$ Dropout(z_1, p=0.5)
16: $z_2 \leftarrow$ GeLU(BN($W_{c2} \cdot z_1 + b_{c2}$)) // Second compression
17: $z_2 \leftarrow$ Dropout(z_2, p=0.3)
18: $y \leftarrow W_o \cdot z_2 + b_o$ // Class logits (Eq. 27)
19: $p \leftarrow$ Softmax(y) // Probability distribution (Eq. 28)
20: $\hat{y} \leftarrow$ argmax(p) // Class prediction (Eq. 29)
21: return $\hat{y}$, p, γ, α, F_{fused}
End

Algorithm 1. Complete TG-CAVNet Forward Pass (End-to-End)

Transformer output is a collection of contextualized token embeddings given by,

$$E_t = \text{BioClinicalBERT}(T), \quad E_t \in R^{N \times d_t} \tag{3}$$

where d_t is the embedding dimensionality. Each embedding vector in E_t encode the local lexicon and global clinical context. Pooling contextual token embeddings creates a concise semantic representation of clinical descriptions. This work computes global text embedding as

$$t = \text{Pool}\,(E_t) \tag{4}$$

where Pool (.) refers to the embedding of the special [CLS] token across token embeddings. Clinical text semantics are captured using this pooling vector.

[CLS] token embedding from the final transformer layer as the global text representation is trained during pre-training to capture sentence-level semantics. Finally, the global text embedding is projected onto a semantic feature space for multimodal visual interactions. The definition of this projection is as follows:

$$t_s = W_t t + b_t \tag{5}$$

where b_t and W_t are the learnable parameters. The clinical semantic representation for text-guided gating feature modulation and text-queried cross-attention is semantic feature vector t_s. TG-CAVNet converts clinical narratives into structured semantic embeddings for multimodal reasoning and automated skin lesion diagnosis. The algorithm of the proposed TG-CAVNet is given in Algorithm 1.

3.2.2. Dermoscopic Image Processing with EfficientNetB4

Dermoscopy provides lesion morphology, pigmentation patterns, texture and structural anomalies for automated skin lesion analysis. TG-CAVNet uses dermoscopic image processing to extract multi-scale and discriminative visual representations while reducing background artefacts and imaging noise. I represents each dermoscopic image that is fed to the model.

To ensure visual backbone compatibility, the input images were preprocessed before feature extraction. Using channel-wise mean and variance normalization, resizing of each image I to a fixed resolution is performed. Data augmentation methods, including random horizontal flipping, rotation and colour perturbation improves resilience and generalization during training. Eq. (6) represents dermoscopic image I with height H and width W and Eq. (7) represents the preprocessed image $\check{I}$.

$$I \in R^{H \times W \times 3} \tag{6}$$

$$\check{I} = \text{Preprocess}\ (I) \tag{7}$$

The EfficientNet-B4 convolutional neural network processes the normalized image $\check{I}$ as the visual backbone. EfficientNet-B4 balances the network depth, width, and input resolution with compound scaling to extract high-level visual information. The backbone creates a comprehensive visual feature map by,

$$F_v = \text{EfficientNetB4}(\check{I}), \qquad F_v \in R^{H^{'} \times W^{'} \times C} \tag{8}$$

where $H^{'}$ and $W^{'}$ are the spatial dimensions of the feature map and C is the feature channel count. TG-CAVNet uses a Feature Pyramid Network (FPN) on the EfficientNet-B4 backbone to record lesion features at several spatial resolutions. The FPN uses lateral connections and top-down upsampling to create feature pyramids at three scales (1/8, 1/16 and 1/32 of the original resolution) from EfficientNet-B4 stages 3, 4 and 5. The multi-scale representation captures fine-grained texture features and large structural patterns. The FPN creates multi-scale feature maps from convolutional stage features using lateral connections and top-down pathways as follows:

$$\{F_v^{(l)}\} = \text{FPN}\ (F_v), \qquad l = \{1,2, \ldots. L\} \tag{9}$$

where l is the pyramid level and L is the scale count. The model can capture the global lesion context and fine-grained local features using this multiscale representation.

Spatial attention is applied to aggregated visual information to improve the lesion-focused feature learning. Spatial attention emphasizes the lesion features and suppresses extraneous backgrounds. Prior to convolution and sigmoid activation, average and max pooling along the channel dimension compute the attention map as follows:

$$A_s = \sigma\,(\mathrm{Conv}([\mathrm{AvgPool}(F_v),\ \mathrm{MaxPool}(F_v)])) \tag{10}$$

where $A_s \in R^{H' \times W'}$. Element-wise multiplication applies the attention map to the visual features as follows:

$$F_v^{sa} = A_s \odot F_v \tag{11}$$

Algorithm TRAIN_TG_CAVNet(D_{train}, Θ, epochs)
Input: Training dataset $D_{train} = (I_i, T_i, y_i)_{i=1}^{N}$
Model parameters Θ (initialized)
Number of training epochs
Output: Optimized parameters Θ*

Initialize AdamW optimizer: lr=2e-5, β_1=0.9, β_2=0.999, *weight_decay*=1e-4
Initialize CosineAnnealingLR scheduler with warmup=300 steps
for epoch = 1 to epochs do:
    Shuffle(D_{train}) // Randomize training order
    for each batch B = (I_b, T_b, y_b) in D_{train}
     do:
        $\hat{y}_b, p_b, \gamma_b, \alpha_b, F_{fused_b}$ ← TG_CAVNet_FORWARD(I_b, T_b)
        // Compute hybrid loss
        L_{focal} ← FocalLoss(p_b, y_b, γ_f=2.0) // Focus on hard examples
        L_{CE} ← CrossEntropyLoss(p_b, y_b) with label smoothing ε=0.1
        L_{Total} ← 0.7·L_{focal} + 0.3·L_{CE}
        // Backward pass with regularization
        ∇Θ ← Backward(L_{Total}) // Compute gradients
        ∇Θ ← ClipGradientNorm(∇Θ, *max_norm*=1.0) // Gradient clipping
        Optimizer.step(∇Θ) // Update parameters
        Scheduler.step() // Update learning rate
    end for
end for
return Θ* // Trained model
End

Algorithm 2. Training with Hybrid Loss (Optimization)

The final visual feature set for the text-guided modulation and cross-attention modules includes the spatially attended features F_v^{sa} and multi-scale representations $\{F_v^{(l)}\}$. EfficientNet-B4, feature pyramid modelling and spatial attention helped TG-CAVNet extract robust multi-scale visual features for multimodal integration and automated skin lesion detection. The algorithm for training with hybrid loss is given in Algorithm 2.

3.3. Text-Guided Gating Feature Modulation

The proposed system uses a text-guided cross-attention mechanism to condition visual feature representations in the semantic textual context to enable successful visual-textual interaction. Visual and textual information are transformed into query, key, and value representations using learnable linear transformations. Textual embeddings generate query vectors, whereas visual embeddings construct key and value matrices. Scaled dot-product attention examines the semantic significance between textual descriptors and visual feature components to calculate attention weights. Although dermoscopic images provide rich visual information, clinical dermatological diagnosis relies heavily on the semantic context of textual descriptions such as lesion colour variation, asymmetry, and border irregularity. When visual cues are modest or unclear, vision-only feature extraction may miss the diagnostically relevant patterns. TG-CAVNet uses clinical semantic information to explicitly adjust visual features using text-guided gating feature modulation, enabling context-aware visual representation learning.
The global semantic embedding t_s of clinical text processing is projected into the visual feature space to correlate its dimensionality with the visual feature channels. This projection is defined as

$$t_p = W_p t_s + b_p \tag{12}$$

where the learnable parameters W_p and b_p plays a vital role. The channel-wise modulation coefficients are conditioned by the projected text representation t_p. The gate generation network maps the projected clinical text embeddings to channel-wise gating coefficients using two fully connected layers. First layer changes text embedding from dimensionality d_t to 1024 using ReLU activation and dropout with a probability of 0.3 for regularization. A second layer projects the representation to C channels, resulting in sigmoid activation and gating values in [0,1].

$$\gamma = \sigma(W_g t_g + b_g), \qquad \gamma \in R^C \tag{13}$$

This design nonlinearly converts clinical meanings into channel-specific visual modulation signals. By applying gating coefficients γ to the spatially attended visual feature map F_v^{sa}, the visual features are recalibrated through channel-wise multiplication given by,

$$F_v^{gate} = \gamma \odot F_v^{sa} \tag{14}$$

This text-conditioned channel attention mechanism weights the visual channels according to the clinical semantic context. We amplified the diagnostically crucial channels and suppressed the less relevant ones. According to the clinical text, the gated visual features F_v^{gate} emphasize lesion characteristics, creating a semantically informed visual representation. In multi-stream fusion, these features complement spatially attended visual features and are combined with cross-attended representations. TG-CAVNet utilizes text-guided gating to bridge the semantic and visual modalities for clinically grounded feature recalibration, which improves discriminative performance and interpretability.

Channel-wise gating enhances feature discrimination by conditioning visual channel weights with clinical semantic information that is absent from visual features. A vision-only attention mechanism can only recalibrate channels based on image statistics, but the text-guided gate knows which lesion attributes are diagnostically relevant for the case, allowing it to amplify channels encoding those patterns and suppress channels responding to irrelevant background or artifact features. This creates a more tailored and therapeutically useful visual representation than the self-attention of picture data alone. Owing to the sigmoid activation used to generate the gating weights, all values must be between zero and one, and the gate network is initialized with near-zero weights; therefore, all channels start training with gating values close to 0.5, preventing excessive channel suppression. The dropout of the gate network during training prevents any clinical attribute in the text from consistently driving a channel toward zero across all samples, keeping all visual channels active and informative throughout learning. The algorithm for gating is given in Algorithm 3.

Algorithm CONCEPT_TO_GATE(t_s, F_v):
Input: Text semantic embedding $t_s \in R^{d_s}$, $[d_s = 512]$
Visual features $F_v \in R^{C\times H\times W}$, $[C = 1792, H = 48, W = 48]$
Output: Gated visual features $F_v^{gate} \in R^{(C\times H\times W)}$,
Channel-wise gating weights $\gamma \in [0,1]^C$

1: // Step 1: Project text to visual feature space
2: $t_p \leftarrow W_p t_s + b_p$ where $t_p \in R^C$, $W_p \in R^{(C\times d_s)}$, $b_p \in R^C$
3: // Step 2: Generate channel-wise gating weights (Gate Generation Network)
4: $h_1 \leftarrow \mathrm{ReLU}(W_{g1} \cdot t_p + b_{g1})$ // Hidden layer: C → 1024
5: $h_1 \leftarrow \mathrm{Dropout}(h_1, p=0.3)$
6: $h_2 \leftarrow W_{g2} \cdot h_1 + b_{g2}$ // Output layer: 1024 → C
7: $\gamma \leftarrow \mathrm{Sigmoid}(h_2)$ // $\gamma \in [0,1]^C$

// Interpretation: γ[c] = clinical_relevance(*visual_channel* ***c*** | *text_description*)
8: // Step 3: Apply channel-wise modulation
9: for c = 1 to C do: // Loop over 1792 channels
10: $F_v^{gate}[c,:,:] \leftarrow \gamma[c] \odot F_v[c,:,:]$ // Channel-wise scaling
11: end for

12: return F_v^{gate}, γ
End

Algorithm 3. Text Guided Gate Feature Modulation

3.4. Text-Queried Cross-Attention Module

Text-guided gating allows global channel-wise visual feature recalibration; however, dermatological diagnosis sometimes requires fine-grained spatial reasoning using clinical descriptors. The text-query cross-attention module in TG-CAVNet allows clinical text embeddings to directly query spatial visual feature maps, enabling semantic alignment between textual descriptors and image regions. The cross-attention technique depends on the question representation and the projected clinical semantic embedding t_p. The query vector is formalized as,

$$Q = W_Q t_p \tag{15}$$

where W_Q is a learnable projection matrix. This formulation ensures that clinical semantics, rather than visual similarity, drive attention. The [CLS] token is a global query designed for lesion-level classification, where clinical text provides a holistic diagnostic impression, rather than spatially anchored word-level descriptions. The [CLS] embedding captures the semantic intent of the full clinical narrative, which is sufficient for identifying globally relevant lesion regions. Token-level queries can theoretically provide deeper, attribute-specific geographical advice, but they are costly in proportion to sequence length and introduce attention diffusion risk when simultaneous token queries provide contradictory spatial signals. The single global query strategy provides sufficient representational richness for binary lesion classification without the instability of multi-token querying because the eight attention heads already diversify the query into eight concurrent semantic subspaces. Future work may extend this to token-level or multi-granular queries for fine-grained segmentation of lesions.

The spatial attention module uses the highest-resolution FPN feature map (1/8 scale) to suppress background artefacts and non-informative areas and focuses on computing efforts on lesion spots. The spatially attended visual feature map F_v^{sa} generates key and value representation. In particular, linear projections yield.

$$K = W_K F_v^{sa} \tag{16}$$

$$V = W_V F_v^{sa} \tag{17}$$

where W_K and W_V are also learnable. The keys indicate spatial visual patterns, whereas the values reflect aggregated visual material. Clinical dermatological descriptions consistently encode eight diagnostic attribute categories, including colour, border, asymmetry, location, size, texture, vascular patterns, and dermoscopic structures, which inspired the choice of eight attention heads in this study. One attention head per attribute can independently focus on visual regions related to the therapeutic notion. By matching the number of heads to semantically distinct attribute types, each head is encouraged to focus on image regions important to one diagnostic idea, encompassing the entire clinical context without any repetition. Utilizing fewer heads would compel numerous unrelated semantic ideas to share a single attention subspace, limiting alignment precision, whereas utilizing many heads would introduce redundant subspaces without useful diagnostic coverage, increasing computation. Thus, the choice of eight aligns the architectural capability of the attention mechanism with the semantic complexity of dermatological clinical language.

Multi-head cross-attention allows TG-CAVNet to attend to several semantic visual subspaces simultaneously and improve its representational capacity. The scaled dot-product attention for each attention head h is calculated as,

$$\mathrm{Attn}_h(Q,K,V) = \mathrm{softmax}\left(\frac{Q_h K_h{}^T}{\sqrt{d_k}}\right)V_h \quad (18)$$

where dk is the key vector dimensionality. All head outputs are concatenated and linearly processed to obtain the aggregated attention output as below.

$$\mathrm{Attn}\,(Q,K,V) = \mathrm{Concat}(\mathrm{Attn}_1,\mathrm{Attn}_2,\ldots\ldots\ldots\mathrm{Attn}_H)W_o \quad (19)$$

where W_o is a learnable projection matrix. The attention output is a contextual aggregate of visual elements weighted by clinical text query relevance. The residual connection and layer normalization retain the visual representation and ensure steady optimization.

$$F_v^{att} = \mathrm{LN}(F_v^{sa} + \mathrm{Attn}(Q,K,V)) \quad (20)$$

Algorithm: TEXT_QUERIED_CROSS_ATTENTION(t_p, F_v)
Input: Projected text embedding $t_p \in R^C$,
Visual features $F_v \in R^{(C\times H\times W)}$
Output: Cross-attended features $F_v^{att} \in R^{(C\times H\times W)}$
1: // Step 1: Flatten and project visual features
2: $F_{flat} \leftarrow$ Reshape(F_v , [C, H×W]) // $R^{(C\times 2304)}$
3: // Step 2: Query-Key-Value projections
4: Q ← $W_Q \cdot t_p$ //Query from text
5: K ← $W_K \cdot F_{flat}$ //Keys from visual
6: V ← $W_V \cdot F_{flat}$ // Values from visual

// $W_Q, W_K, W_V \in R^{(d\times C)}$, d=64 per head (total 512)
7: // Step 3: Multi-head attention (8 heads)
8: $d_k \leftarrow 64$ // Dimension per head
9: for h = 1 to 8 do:
10: // Split into heads
11: $Q_h \leftarrow Q[(h\text{-}1)\times d_k : h\times d_k]$ // R^{64}
12: $K_h \leftarrow K[(h\text{-}1)\times d_k : h\times d_k, :]$ // $R^{(64\times 2304)}$
13: $V_h \leftarrow V[(h\text{-}1)\times d_k : h\times d_k, :]$ // $R^{(64\times 2304)}$
// Scaled dot-product attention
14: $S_h \leftarrow (Q_h \cdot K_h)/\sqrt{d_k}$ // Similarity scores
15: $A_h \leftarrow \mathrm{Softmax}(S_h)$ // Attention weights $R^{(2304)}$
16: $O_h \leftarrow A_h \cdot V_h^T$ // Head output R^{64}
17: end for
18: // Step 4: Combine heads and project
19: $\mathcal{O} \leftarrow \mathrm{Concat}(O_1, O_2, \ldots, O_8)$ // R^{512}
20: $O_{proj} \leftarrow \mathcal{O} \cdot W_o + b_o$ // Eq. (19): R^C
21: // Step 5: Residual connection and normalization
22: $O_{reshaped} \leftarrow \mathrm{Reshape}(O_{proj}, [C, H, W])$ // Match spatial dimensions
23: $F_v^{att} \leftarrow \mathrm{LayerNorm}(F_v + O_{reshaped})$ // Eq. (20)
24: return F_v^{att}
End

Algorithm 4. Text-Queried Cross-Attention (Semantic-Spatial Alignment)

The final output F_v^{att}, is a text-attended visual feature map that emphasizes clinically relevant spatial locations. This format captures fine-grained semantic–visual correlations and complements the globally modulated gated features obtained from Eq. (14). These methods allow TG-CAVNet to execute clinically grounded visual reasoning that mimics human diagnostic behaviour and improves interpretability using text-driven attention mapping. The algorithm for text queried cross attention is given in Algorithm 4.

3.5. Multi-Stream Visual Feature Fusion

A unified multimodal embedding is created from the feature representations after cross-attention interaction and modulation. The cross-attention output shows contextual congruence between the text and visual modalities, whereas the modulated visual representation preserves text-influenced visual signals. The integration of these complementary representations creates a fused feature representation F_{fused} which is used for classification. This fused representation allows the network to capture richer multimodal relationships between textual descriptors and dermoscopic image features using contextual attention-based interaction and feature modulation.

The text-guided gating and text-queried cross-attention modules produce complementary visual representations of global channel-wise semantic significance and fine-grained spatially localized semantic alignment. Each stream improves visual understanding, but using a single representation may not model the lesion features. TG-CAVNet uses a multi-stream visual feature fusion module to adaptively combine both representations for robust and semantically coherent visual reasoning. Text-guided channel modulation produces gated visual features F_v^{gate} and the text-queried cross-attention mechanism produces cross-attended visual features F_v^{att}, which are received by the fusion module. Direct fusion can cause a scale imbalance because these feature streams originate from various modulation processes and have different magnitudes. L2 normalization was applied separately to each stream for feature alignment and numerical stability as given in Eq. (21) and Eq. (22).

$$\hat{F}_v^{gate} = \frac{F_v^{gate}}{||F_v^{gate}||_2} \quad (21)$$

$$\hat{F}_v^{att} = \frac{F_v^{att}}{||F_v^{att}||_2} \quad (22)$$

This normalization step aligns the feature distributions so that neither stream dominates the fusion owing to scale disparities. TG-CAVNet uses adaptive fusion gating to dynamically control the stream contributions. The normalized feature representations were concatenated and passed through a learnable gating network.

$$\alpha = \sigma\left(W_f\left[\hat{F}_v^{gate}, \hat{F}_v^{att}\right] + b_f\right) \quad (23)$$

The fusion weight learned for each sample is $\alpha \in [0,1]$ and the gating parameters W_f and b_f are the learnable fusion weights that dynamically balances channel-wise modulated features and spatially attended features respectively. Based on lesion complexity and textual context, this method allows the network to prioritize global semantic recalibration over local semantic alignment. The normalized weighted combination of the two streams produces the fused visual representation as follows.

$$F_{fused} = \alpha.\hat{F}_v^{gate} + (1 - \alpha).\hat{F}_v^{att} \quad (24)$$

Algorithm: ADAPTIVE_MULTI_STREAM_FUSION (F_v^{gate}, F_v^{att})
Input: Gated features $F_v^{gate} \in R^C$,

Cross-attended features $F_v^{att} \in R^C$
Output: Fused representation $F_{fused} \in R^C$,
Fusion weight $\alpha \in [0,1]$
1: // Step 1: L2 normalization for scale alignment
2: $norm_{gate} \leftarrow ||F_v^{gate}||_2$ // L2 norm
3: $norm_{att} \leftarrow ||F_v^{att}||_2$
4: $\hat{F}_v^{gate} \leftarrow F_v^{gate}/(norm_{gate} + \varepsilon)$ // ε=1e-8
5: $\hat{F}_v^{att} \leftarrow F_v^{att}/(norm_{att} + \varepsilon)$
6: // Step 2: Generate adaptive fusion weight
7: $f_{concat} \leftarrow$ Concat($\hat{F}_v^{gate}$, $\hat{F}_v^{att}$) // R^{2C}
8: $\alpha_{logits} \leftarrow W_f \cdot f_{concat} + b_f$ // Linear projection $W_f \in R^{(1\times 2C)}$,$b_f \in R$
9: $\alpha \leftarrow$ Sigmoid(α_{logits}) // $\alpha \in [0,1]$
10: // Step 3: Weighted combination
11: $F_{fused} \leftarrow \alpha \cdot \hat{F}_v^{gate} + (1-\alpha) \cdot \hat{F}_v^{att}$
12: return F_{fused} , α
End

Algorithm 5. Adaptive Multi-Stream Fusion (Dynamic Integration)

Two qualities make this fusion approach stable. Before fusion, L2 normalization ensures that neither stream dominates the fused representation owing to the feature magnitude. After normalization, both streams enter the fusion at equal scales, allowing the learned fusion weight α to determine their relative contribution based on content rather than scale. Owing to the sigmoid activation of α, its value is limited to zero to one for any finite input. This ensures that both streams contribute positively and are never entirely quiet, regardless of the input. The representation does not collapse into one pathway, which would reduce the multistream architecture to a single-stream model and remove the complementary information carried by the second stream. By adding a small constant ε to the normalization denominator, numerical stability is maintained when the feature norms approach zero, minimizing division instability during early training. These properties ensure that the fused representation always contains a balanced, scale-normalized combination of channel-level and spatial-level semantic information, as shown in Table 8, which shows consistent performance gains when both streams are retained compared with the single-stream ablation.

Instead of static concatenation or averaging, this approach blends characteristics that encode complementing semantic-visual information to model cross-modal interactions.
The fusion of global and local text-conditioned visual cues creates a compact and semantically enriched feature vector F_{fused}. TG-CAVNet uses synergistic information from several text-guided visual reasoning pathways while maintaining interpretability and robustness using this fused representation as the final visual embedding of the classification

head. The algorithm for Adaptive Multi-Stream Fusion is given in Algorithm 5.

3.6. Classification Head

The classification head of TG-CAVNet maps the unified multimodal representation to lesion class predictions as the final decision-maker. The classification head compresses, stabilizes, and discriminates while minimizing overfitting using the multi-stream fusion module's fused visual representation F_{fused}.This module predicts the lesion class and confidence scores. The fused feature vector undergoes a learnable linear projection, batch normalization, and nonlinear activation to reduce the feature dimensionality and improve discrimination. Definition of the transformation.

$$z = \mathrm{GeLU}\left(\mathrm{BN}(W_c F_{fused} + b_c)\right) \tag{25}$$

where W_c and b_c are the learnable parameters of the compression layer. Gradient flow and model expressiveness are improved by batch normalization, which stabilizes the feature distribution during training and Gaussian Error Linear Unit (GeLU) activation smoothing non-linearity. The compressed feature representation is regularized with dropout to reduce overfitting and improve generalization.

$$z^{'} = \mathrm{Dropout}(z) \tag{26}$$

where randomly deactivated units inhibit the training feature co-adaptation. The final classification layer generates class logits y from the regularized feature vector $z^{'}$ as follows:

$$y = W_o z^{'} + b_o \tag{27}$$

where W_o and b_o are learnable parameters and $y \in R^K$, K denotes the lesion class count. Logits are processed using softmax to obtain normalized class probabilities as follows.

$$p = \mathrm{softmax}(y) \tag{28}$$

where each element pk indicates the estimated class k input lesion probability. Class prediction is finalized as given in Eq. (29) associated with benign or malignant disease. Calibrated confidence scores from probability vector p enable clinically useful decision-making support.

$$\hat{y} = \underset{k}{\mathrm{argmax}}\, p_k \tag{29}$$

TG-CAVNet completes the automated skin lesion identification pipeline by converting the fused multimodal representation into accurate and reliable lesion prediction using this classification head.

Textual descriptors and dermoscopic images reflects the diagnostic procedure of clinical dermatology, which combines visual inspection with contextual clinical information. The model dynamically associates visual lesion structures with semantic descriptors using a text-guided cross-attention technique, increasing the interpretability and discrimination of the learned representation. Visual feature modulation also emphasizes clinically significant patterns by conditioning visual features in the textual context. These components strengthen multimodal representation learning and improve the accuracy of automated skin lesion detection.

4. Experimental Setup

4.1. Dataset Description

To assess the proposed TG-CAVNet framework, the Multimodal Dermoscopic Dataset from IEEE Dataport was used. This dataset supports multimodal learning for automated skin lesion analysis using aligned dermoscopic images and clinical text description. A description of the dataset is given in table 3. The collection includes 6,194 lesion instances with dermoscopic images, clinical text descriptions and ground truth diagnosis labels. A strict patient-independent evaluation process was ensured by splitting the dataset into 3,904 training samples, 690 validation samples, and 1,600 test samples with no overlap. The training set comprised 2,679 benign and 1,225 malignant cases, and the validation set comprised 473 benign and 217 malignant samples. Similar to the clinical dermatology datasets, the test set included 1,022 benign and 578 malignant lesions.

Table 3. Dataset Description

	Total images	**Benign**	**Malignant**
Training Set	3904	2679	1225
Validation Set	690	473	217
Testing Set	1600	1022	578
Total	6194	4174	1500

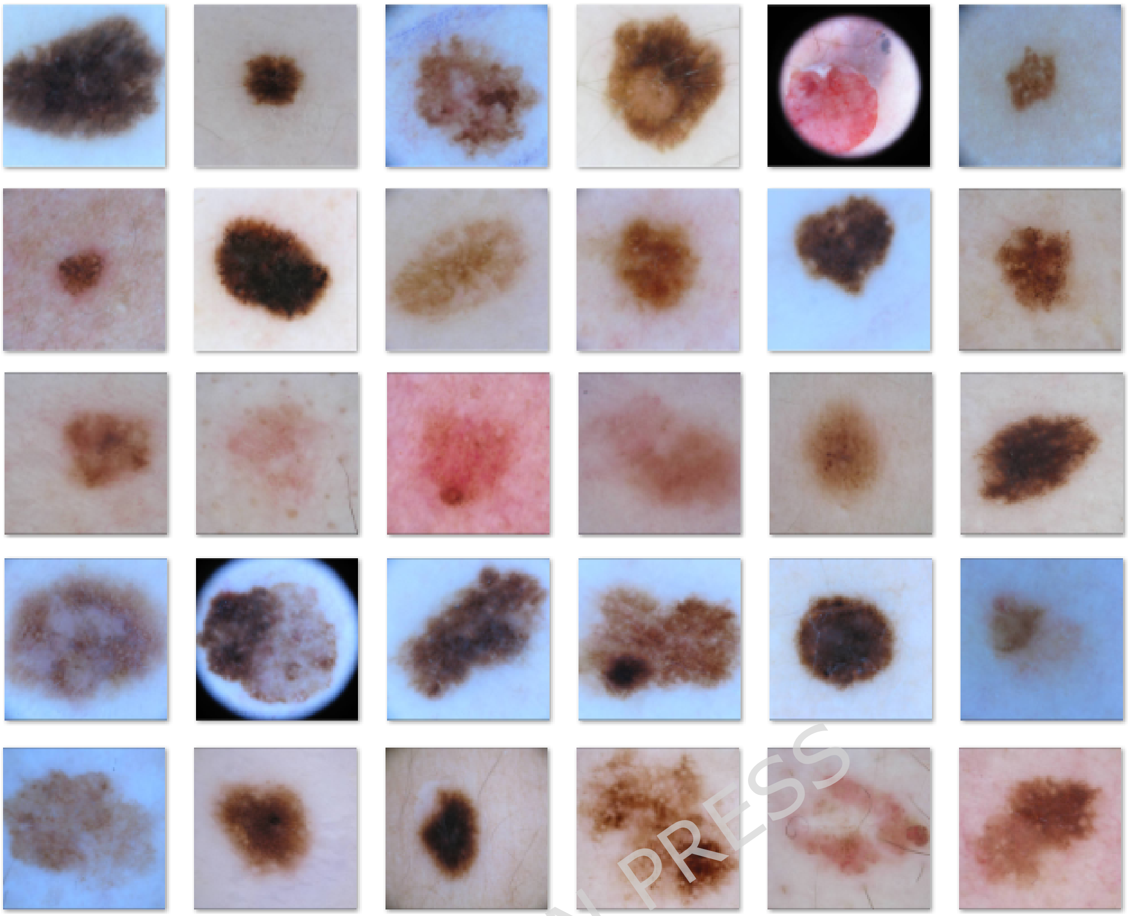


Figure 3. Sample Image Dataset

Figure 3 shows a sample of lesion images in the dataset and figure 4 shows a sample of free text in the dataset. Dermoscopy images were scaled and standardized to suit the EfficientNet-B4 backbone input requirements and typical data augmentation improved generalization during training. Clinical text descriptions were tokenized and encoded using Bio-ClinicalBERT to preserve domain-specific semantics. This dataset is ideal for testing the text-guided feature modulation and cross-attentive multimodal learning mechanisms of TG-CAVNet because of its well aligned image–text pairs and well curated non-overlapping splits.

Figure 4. Sample Text Dataset

By deep learning standards, 6,194 samples are few; however, transfer learning reduces the sample requirement. EfficientNet-B4 and Bio-ClinicalBERT, pretrained on over one million ImageNet images and millions of clinical notes from the MIMIC-III corpus, respectively, provided rich and generalizable representations in their weights. Only the final layers of each encoder were fine-tuned, so the model learned a few task-specific parameters from the dermoscopic dataset instead of starting from scratch. Figure 11 shows a minor discrepancy between the training accuracy (98.51%) and validation accuracy (98.55%), demonstrating that the model generalizes well without overfitting, despite the dataset size. By regularizing the learning process and preventing majority class patterns, data augmentation, inverse frequency class weighting, and label smoothing enhance sample usage.

4.2. System Implementation

The proposed TG-CAVNet uses a multimodal learning framework to integrate dermoscopic images and clinical text. Table 4 summarizes all the key hyperparameters. The dermoscopic images were reduced to 384 × 384 pixels and normalized using ImageNet statistics. In all studies, WordPiece tokenized clinical text descriptions with a maximum sequence length of 128 and a batch size of eight. A fixed random seed was used to ensure reproducibility. EfficientNet-B4 extracted 1792-dimensional visual features, whereas Bio-ClinicalBERT encoded clinical text with fine-tuned final four transformer layers. The AdamW optimizer, cosine annealing learning rate schedule, gradient clipping, and proximal regularization were used for model training. Class imbalance was addressed using label smoothing, inverse-frequency class weighting, and hybrid focal and cross-entropy losses. Training used data augmentation to improve the generalization. Lesion-relevant regions were visualized using Grad-CAM for interpretability.

Table 4. Hyperparameter Description

Category	Parameter	Value
Input	Image resolution	384 × 384 px
	Image normalization (mean, std)	[0.485,0.456,0.406], [0.229,0.224,0.225]
	Text sequence length	128 tokens
Batching & reproducibility	Batch size	8
	Random seed	42
Visual encoder	Backbone	EfficientNet-B4 (pretrained)
	Visual output dim	1792
Text encoder	Model	Bio_ClinicalBERT (fine-tune last 4 layers)
	Hidden dim	768
Cross-modal / gating	Attention heads	8
	Attention dropout	0.1
	Text-guided gating	Sigmoid; MLP 768→1792→1792; gate dropout 0.3
Fusion head	Architecture	1792 → 1024 (ReLU, BN, dropout=0.5) → 512 (ReLU, BN, dropout=0.3) → 2
Optimization	Optimizer	AdamW (β_1=0.9, β_2=0.999)
	Learning rate (initial)	2.0×10^{-5}
	Weight decay	1.0×10^{-4}
	Gradient clipping (max norm)	1.0
	Proximal μ (stabilization)	0.01
Training schedule	Epochs	10
	Warmup steps	300
	LR scheduler	Cosine annealing
Loss / class balancing	Loss	Hybrid: Focal + CE
	Focal γ_f (used)	2.0 (within 1.0–3.0 range)
	Focal weight / CE weight	0.7 / 0.3
	Label smoothing	0.1
	Class weights	Inverse-frequency (auto)
Regularization & augmentation	Data augmentation	H/V flip, rotation, colour jitter
	Dropout (fusion head)	0.5 / 0.3

Explainability	XAI method / layer	Grad-CAM on last conv block; heatmap threshold 0.3

4.3. Evaluation Metrics

Multiple assessment metrics were used to evaluate the proposed TG-CAVNet architecture. These metrics measure the classification accuracy, class-wise discrimination, class imbalance robustness and agreement beyond chance. TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

- **Accuracy:** The percentage of correctly categorized samples is the accuracy. This statistic estimates performance but may be biased in imbalanced datasets.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (30)$$

- **Precision:** Precision measures positive-forecast accuracy as in Eq. (31). Low false-positive rates with high accuracy help reduce unnecessary healthcare procedures.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (31)$$

- **F1 score:** F1 score is the harmonic mean of precision and recall. It fairly evaluates class imbalance.

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision}+\text{Recall}} \quad (32)$$

- **Recall:** Recall evaluates the model accuracy in identifying positive cases. Skin lesion detection requires a strong memory to avoid undetected malignancies.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (33)$$

- **Matthews Correlation Coefficient (MCC):** The MCC, which analyses all four confusion matrix components, is robust and ideal for imbalanced datasets. Values vary from -1 (complete disagreement) to +1 (perfect prediction).

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (34)$$

- **Cohen's Kappa:** Cohen's Kappa accounts for chance agreement between projections and ground truth:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (35)$$

Here, p_o represents the observed agreement and p_e represents the expected agreement by chance. A higher kappa indicates greater reliability.

- **Hamming Loss:** The percentage of incorrect predictions is the Hamming Loss. The reduced values improved the prediction consistency.

$$\text{Hamming Loss} = \frac{FP+FN}{TP+TN+FP+FN} \quad (36)$$

- **Jaccard Score (Intersection over Union):** The Jaccard Score evaluates the predicted true-label overlap.

$$\text{Jaccard Score} = \frac{TP}{TP+FP+FN} \quad (37)$$

- The class-wise Jaccard scores indicate how well the model identifies benign and malignant tumors.

$$Jaccard_{Benign} = \frac{TN}{TN+FP+FN} \quad (38)$$

$$Jaccard_{Malignant} = \frac{TP}{TP+FP+FN} \quad (39)$$

- **AUC-ROC curve:** The ROC curve plots the true positive rate (TPR) against the false positive rate (FPR). AUC-ROC values closer to 1 indicate better model discrimination across all the thresholds.

$$TPR = \frac{TP}{TP+FN} \quad (40)$$

$$FPR = \frac{FP}{FP+TN} \quad (41)$$

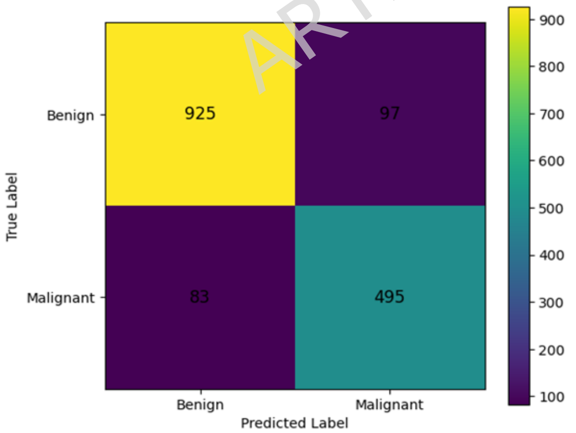
- **PR Curve:** The precision-recall curve compares the Precision and recall at different judgment thresholds. For imbalanced datasets, it highlights the positive (malignant) class performance.

5. Results and Discussion

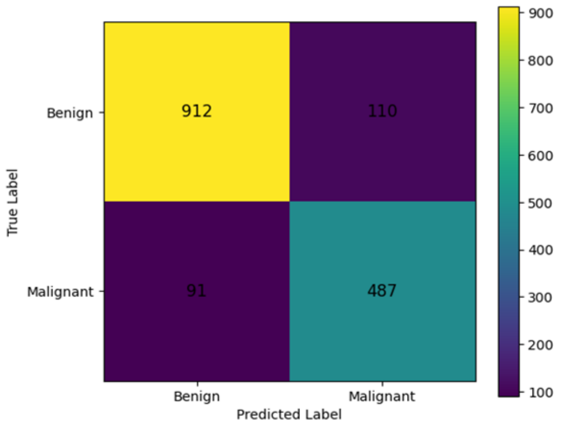
5.1. Quantitative Analysis

The proposed text-guided cross-attentive multimodal architecture was statistically compared with three previous multimodal skin lesion analysis frameworks: MMF-Net [5], Attention-fused multimodal [7], and PanDerm [8]. All models were tested under equal data splits and experimental settings for a fair comparison.

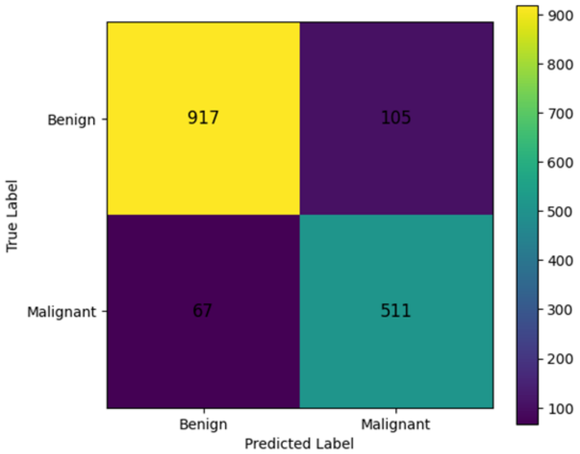
The confusion matrices of the baseline models is depicted in Figure 5 and Figure 6 shows the confusion matrix of the proposed TG-CAVNet, when compared, shows clear performance improvements in benign and malignant categorization. The number of malignant false negatives (81) and benign false positives (67) was significantly lower in TG-CAVNet than in MMF-Net. Similar to the Attention-fused Multimodal baseline, TG-CAVNet reduced the benign-to-malignant misclassification from 110 to 67 and increased the malignant detection consistency. PanDerm had more false positives and negatives than TG-CAVNet, despite better malignancy identification. TG-CAVNet had the truest positives for malignant lesions and the least class confusion, demonstrating higher class separability. These results show that text-guided gating, cross-attention and adaptive fusion processes improve benign-malignant lesion discrimination over multimodal baselines.



a) Confusion matrix - MMF-Net (Baseline)



b) Confusion matrix – Attention fused Multimodal (Baseline)



c) Confusion matrix – PanDerm(Baseline)

Figure 5. Confusion Matrices of Baseline Models

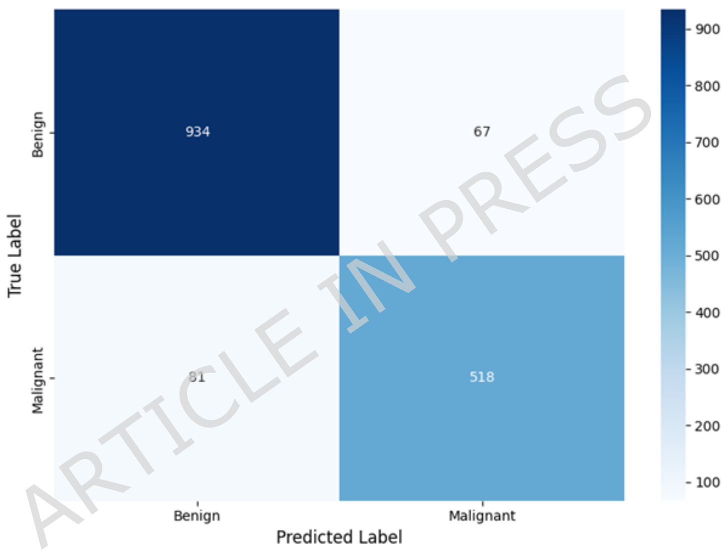


Figure 6. Confusion Matrix of Proposed TG-CAVNet

Figure 7 shows how the text-guided cross-attention mechanism affects the classification accuracy of the proposed TG-CAVNet compared with the three multimodal baseline models. PanDerm (89.26%), MMF-Net (88.73%), and the attention-fused multimodal method (87.47%) were less accurate than TG-CAVNet (90.75%). This enhancement shows that clinical text embeddings can explicitly query and drive spatial visual feature representations without using static or shallow fusion techniques. Text-guided cross-attention improves semantic–visual alignment, lesion discrimination and classification performance.

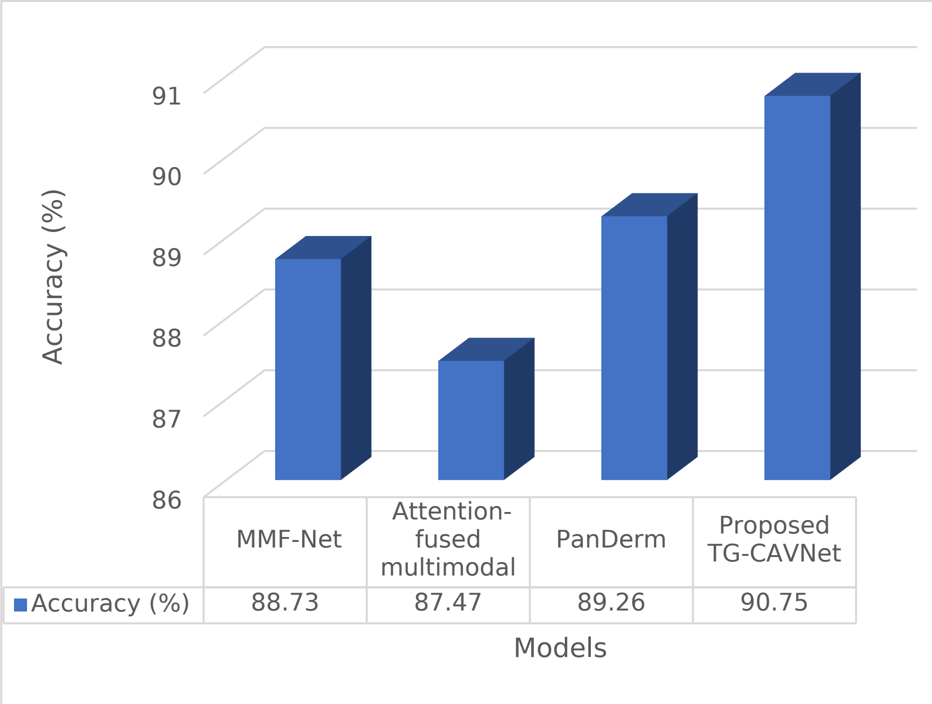


Figure 7. Performance analysis of Proposed TG-CAVNet based on Accuracy

Table 4 shows that TG-CAVNet has the highest precision, recall, F1-score, MCC, and Cohen's kappa of all models, indicating better classification reliability and agreement beyond chance. The significant gains in MCC and Cohen's kappa demonstrate the class-imbalance resilience of TG-CAVNet and its consistent diagnostic performance across both lesion categories.

Table 4. Quantitative analysis of Proposed TG-CAVNet based on classification metrics

Model	Precision	Recall	F1-Score	MCC	Cohen's Kappa
MMF-Net [5]	0.8362	0.8564	0.846	0.744	0.755
Attention-fused multimodal [7]	0.8157	0.8426	0.829	0.724	0.735
PanDerm [8]	0.8295	0.8841	0.856	0.767	0.773
Proposed TG-CAVNet	**0.8855**	**0.8648**	**0.875**	**0.8017**	**0.8016**

Table 5 presents the clinically relevant diagnostic parameters used to assess the TG-CAVNet. A low Hamming Loss (0.0925) suggests few misclassified instances, but a high macro Jaccard score (0.8205) indicates balanced performance across benign and malignant classifications. The suggested text-guided cross-attention and adaptive fusion techniques improve clinically relevant decision-making, as demonstrated by the class-wise Jaccard scores for benign lesion recognition and malignant detection.

Table 5. Quantitative analysis of Proposed TG-CAVNet based on Clinical diagnostic metrics

Clinical Diagnostic Metrics	Values
Hamming Loss	0.0925
Jaccard Score (Macro)	0.8205
Jaccard Benign	0.8632
Jaccard Malignant	0.7778

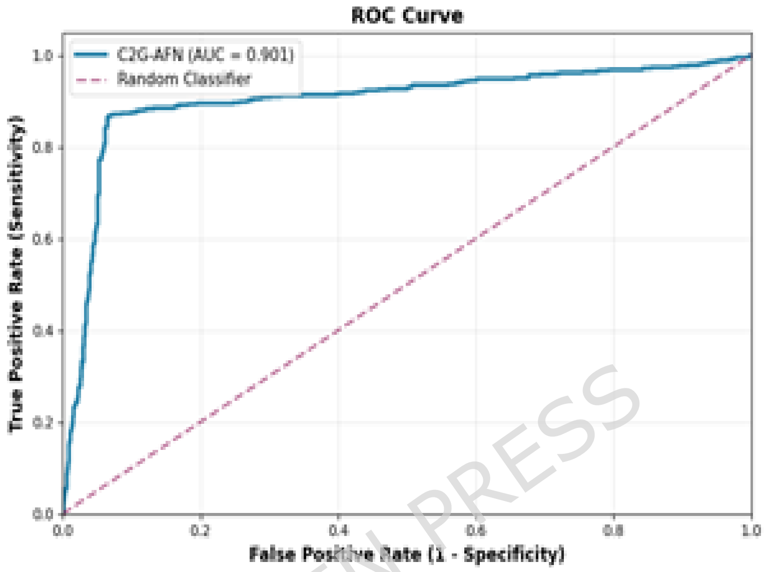


Figure 8. ROC Curve of proposed TG-CAVNet

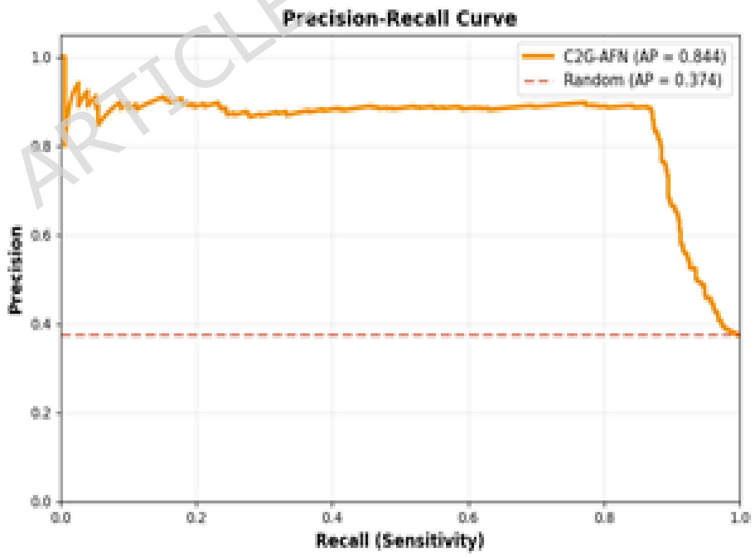


Figure 9. PR Curve of proposed TG-CAVNet

Figures 8 and 9 show the strong discrimination of the TG-CAVNet. The ROC curve in Figure 8 displays a high AUC, demonstrating an efficient distinction between benign and malignant lesions across decision thresholds. The precision–recall curve in Figure 9 shows robust malignant lesion identification with high precision and a wide recall range. These curves show that TG-CAVNet is reliable under class imbalance and is suitable for clinically significant skin lesion categorization.

Figure 10. Training and Validation Loss

The training dynamics and convergence behavior of the proposed TG-CAVNet model are shown in Figures 10 and 11, respectively. Figure 10 shows that the training and validation losses decreased progressively throughout the epochs, with the validation loss reaching 0.0862 at epoch 7, demonstrating steady optimization without overfitting.

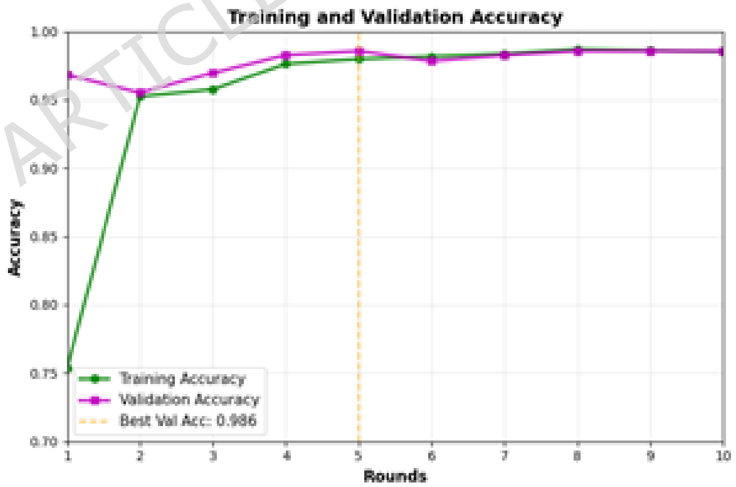


Figure 11. Training and Validation Accuracy

Figure 11 shows rapid convergence, with training and validation accuracies exceeding 95% by epoch 2 and aligning throughout the training. The model's ultimate training and validation accuracies were 0.9851 and 0.9855, respectively, with the best validation accuracy at epoch 5. Early convergence and close matching of the training and validation curves indicate the efficacy of the text-guided cross-attentive multimodal learning technique. By combining visual and written information, the multimodal learning technique increases the representation quality. Dermoscopy images show lesions in detail, but textual descriptors provide contextual semantic information that images cannot. This approach emphasizes diagnostically significant visual structures by dynamically aligning various modalities using cross-attention. Multimodal interaction improves feature discrimination and classification ability, as demonstrated in the experiments.

5.2. Ablation Studies

An ablation study was conducted to examine the effect of visual backbone selection in TG-CAVNet while maintaining all other components. To guarantee a fair comparison, the Bio-ClinicalBERT text encoder, text-guided gating module, cross-attention mechanism, fusion approach, loss function, optimizer, and dataset splits were fixed across all tests.

Table 6. Ablation study on visual backbone selection within TG-CAVNet

Visual Backbone	Accuracy (%)	Precision	Recall	F1-Score	AUC-ROC	Training Time (sec)
TG-CAVNet (Inception-V3)	78.88	0.75	0.63	0.68	0.82	569
TG-CAVNet (ResNet50)	76.38	0.73	0.55	0.63	0.79	357
TG-CAVNet (XceptionNet)	75.94	0.67	0.65	0.66	0.80	589
TG-CAVNet (VGG16)	75.75	0.69	0.60	0.64	0.81	537
TG-CAVNet (DenseNet201)	75.25	0.66	0.65	0.66	0.80	411
TG-CAVNet (EffcientNetB4)	**90.75**	**0.89**	**0.87**	**0.88**	**0.80**	**571**

Table 6 shows that visual backbones have varied performances and computational trade-offs. While lighter designs such as ResNet-50 train faster but have worse recall, richer models such as XceptionNet and DenseNet201 increase recall without balancing the metrics. With a slight increase in training time, EfficientNet-B4 consistently performed best in terms of accuracy, precision, recall, and F1-score. These results support the use of EfficientNet-B4 as the final TG-CAVNet visual backbone. EfficientNet-B4's compound scaling law optimizes the depth, width, and resolution to maximize the representational capacity per parameter, explaining its higher performance. ResNet-50 and other lighter backbones cannot capture fine-grained dermoscopic texture patterns because of their smaller Vapnik Chervonenkis (VC) dimensions. EfficientNet-B4's consistent accuracy–recall balance reveals it has the best bias-variance trade-off among the evaluated topologies.

Table 7 shows an ablation study that examined the effects of clinical text encoder selection on TG-CAVNet while fixing all other components. Generic BERT-based encoders have low recall and F1-score, indicating poor clinical semantic modeling. The T5-based encoder is fast but requires extensive training. The final TG-CAVNet model used Bio-ClinicalBERT as the text encoder because it balances accuracy, recall and computational economy.

Table 7. Ablation Study on Clinical Text Encoder Selection Within TG-CAVNet

Text Encoder	Accuracy (%)	Precision	Recall	F1-score	AUC	Training time (sec)
BERT (ClinicalBERT)	70.44	0.80	0.24	0.37	0.80	346
T5	99.06	0.98	0.99	0.98	0.99	1760
Bio-ClinicalBERT	**90.75**	**0.89**	**0.87**	**0.88**	**0.80**	**571**

Domain-adapted pretraining on MIMIC-III clinical notes aligns Bio-ClinicalBERT's embedding space with the dermatological lexicon, maximizing the conditional mutual information. Because T5's sequence to sequence pretraining target is misaligned with classification, its 99.06 percent accuracy with excessive training time shows encoder-task overfitting rather than generalization. Semantically grounded embeddings provide Bio-ClinicalBERT with the best accuracy efficiency trade-off.

A component-based ablation study was conducted to assess the core modules of TG-CAVNet. As shown in Table 8, eliminating the text-guided gating module lowers the accuracy to 86.12%, emphasizing channel-wise semantic recalibration. Excluding the cross-attention module reduces the performance to 85.34%, emphasizing text-driven spatial alignment. Without the fusion module, the accuracy drops to 83.97%, proving that the adaptive integration of gated and attended characteristics is crucial. The full TG-CAVNet arrangement had the maximum accuracy, verifying the complementary and synergistic contributions of all components.

Table 8. Component based ablation study within TG-CAVNet

Configuration	Gating	Cross-Attn	Fusion	Accuracy (%)
Full TG-CAVNet	□	□	□	**90.75**
Without Gating	□	□	□	86.12

Without Cross-Attention	□	□	□	85.34
Without Fusion	□	□	□	83.97

Eliminating the gating module reduces text-conditioned channel recalibration, resulting in a 4.63% loss of conditional mutual information. The larger 5.41% decline from removing cross-attention demonstrates the increased impact of spatial semantic alignment on lesion boundary discrimination. The 6.78% decline without the fusion module, the largest of the three, is consistent with ensemble theory which involves integrating two decorrelated streams (channel-level and spatial-level) reduces the feature variance and maximizes the individual performance.

5.3. Interpretability and Visualization

The t-SNE visualization of the learned feature embeddings throughout the training, validation, and test splits is shown in Figure 12. A well-distributed and overlapping structure among the three divisions indicates consistent feature representations and no data leakage or split bias. TG-CAVNet develops stable and discriminative embeddings that generalize beyond the training data, as clear grouping patterns between benign and malignant samples persist across all splits. This division reinforces the resilience of the learned multimodal representations and quantitative outcomes.

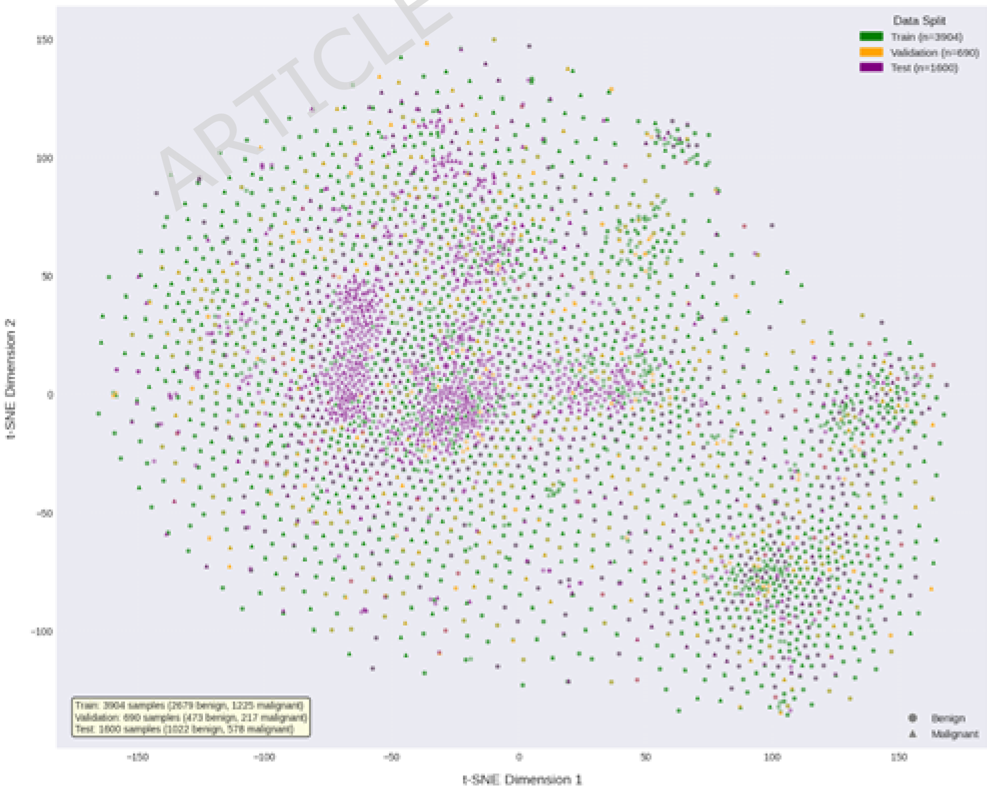


Figure 12. t-SNE Visualization by Data Split

An improved t-SNE analysis of the learned feature space shows the class-wise distribution and separability in the dermoscopic dataset in Figure 13. Because many lesion types appear identical, kernel density estimation shows partially overlapping but distinct density regions for benign and malignant samples. Despite some overlap, malignant samples tended to occupy more compact and structured embedding space regions, as shown by the convex hull boundaries. These results show that TG-CAVNet learns discriminative and clinically realistic representations to separate classes while allowing dermoscopic variability.

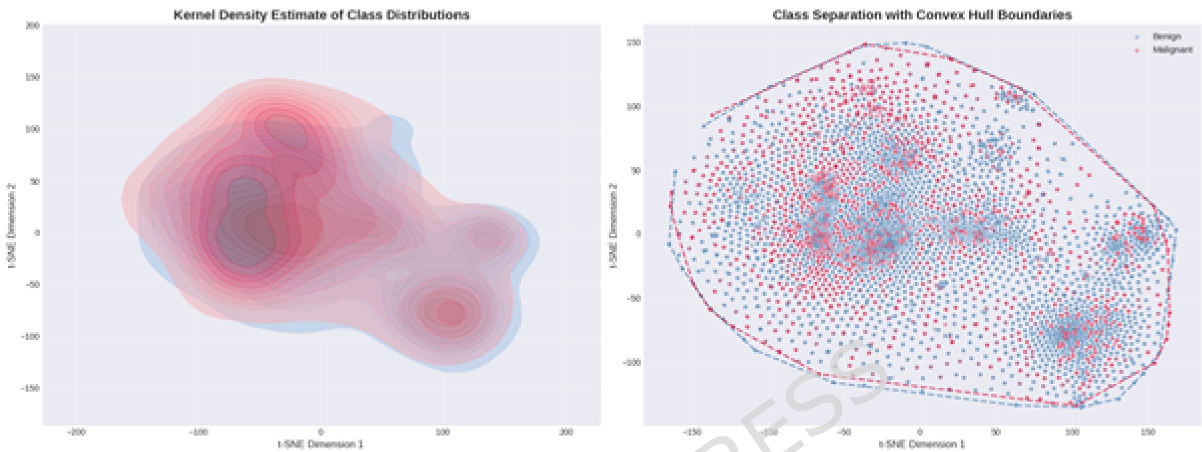


Figure 13. Advanced t-SNE analysis of Dermoscopic dataset

Figures 12 and 13 show that the learned multimodal embeddings form distinct, separable clusters for benign and malignant classes across the training, validation, and test splits, confirming that the joint image - text representations encode diagnostically discriminative information rather than modality-specific noise. The constancy of this separation across all three splits implies that the representations generalize outside the training distribution, which is necessary for clinical classification and identifiability.

Figure 14 shows the interpretability of TG-CAVNet for a benign lesion instance using enhanced text-guided explainable AI (XAI). The clinical language presents mixed lesion characteristics, including comforting brown and light-brown hues and worrisome rough textures and darker brown patches, resulting in a balanced semantic influence with a net text contribution of zero. Despite these conflicting textual indications, TG-CAVNet predicted a benign lesion with a high confidence score of 0.928 and a low malignant chance of 0.072.

The text-guided attention map showed that the model relied on strong visual evidence to focus on lesion sites with benign morphology in the absence of clear textual risk indications. This example shows how TG-CAVNet integrates clinical text without amplifying ambiguous cues, maintaining robust image-driven judgments and transparent, clinically interpretable reasoning.

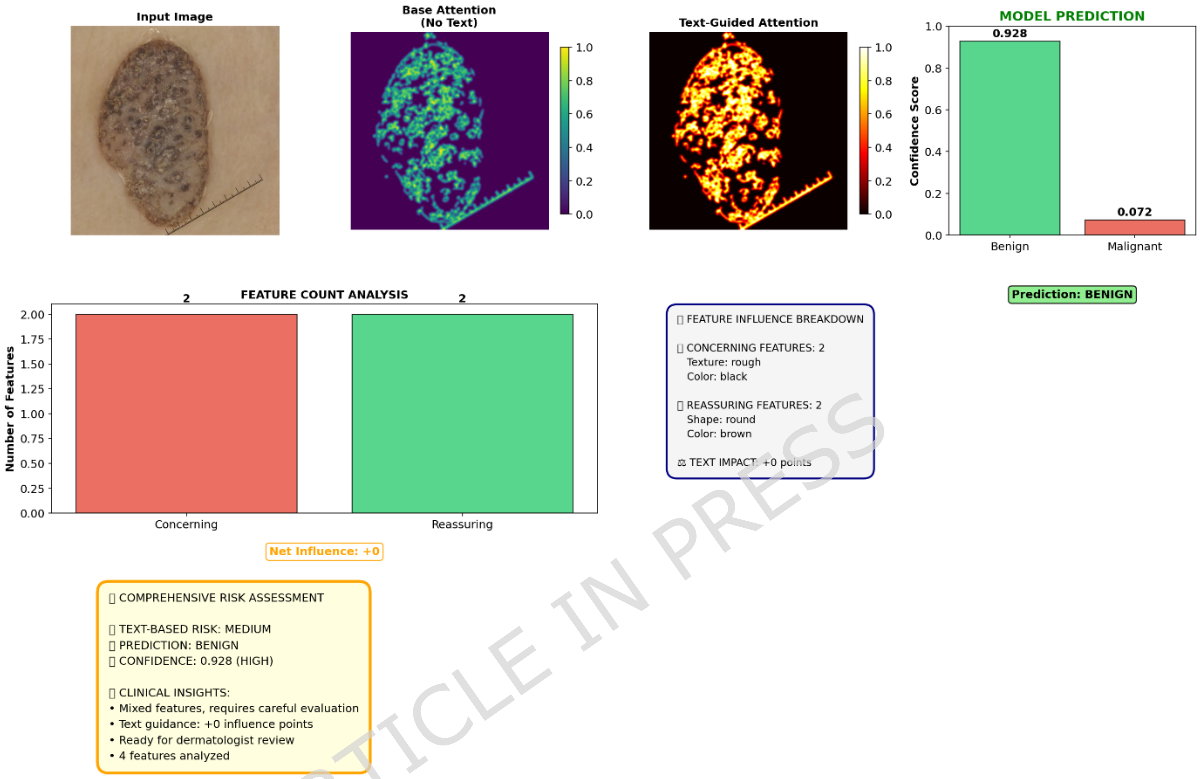


Figure 14. Explainable text-guided attention and feature attribution for benign lesion classification

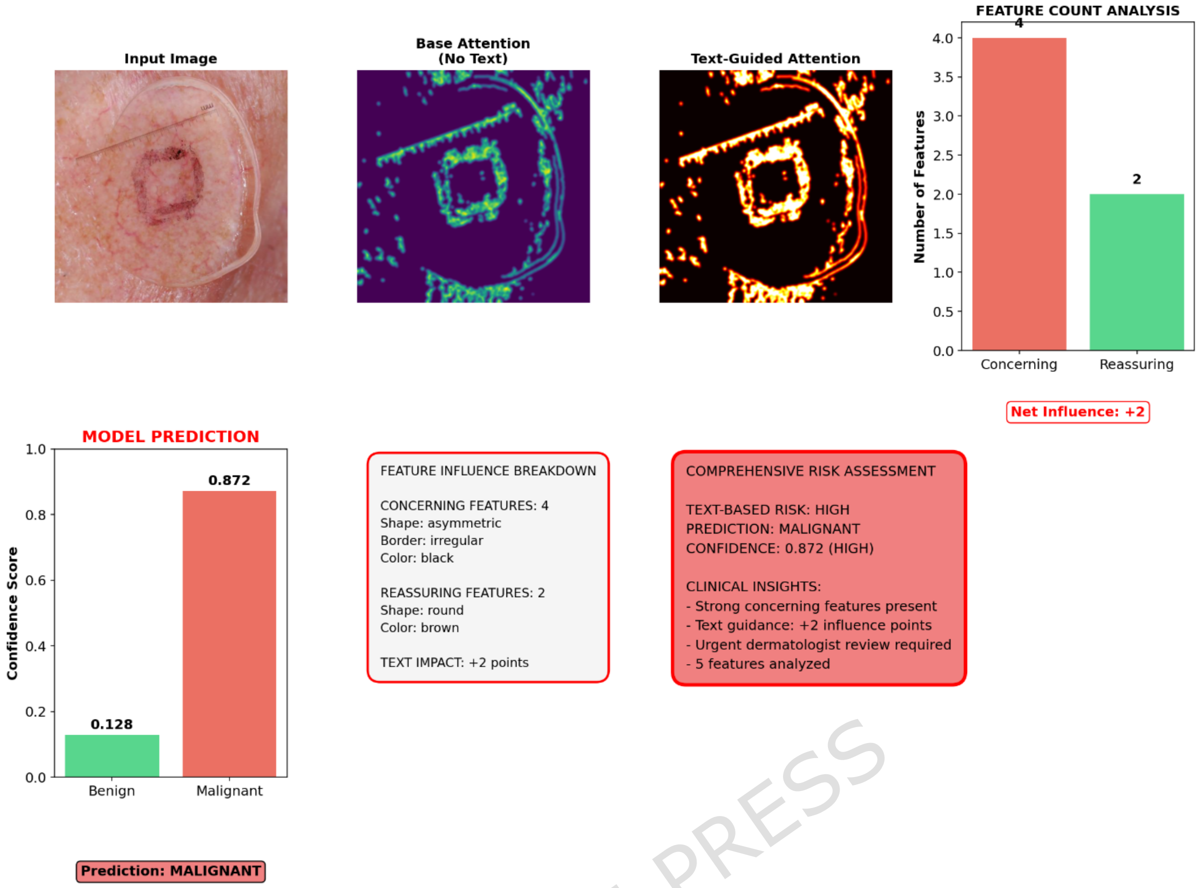


Figure 15. Explainable text-guided attention and feature attribution for malignant lesion classification

Figure 15 shows the TG-CAVNet interpretability analysis for a malignant lesion case, demonstrating how text-guided attention affects high-risk decision-making. In this case, the model predicted malignancy with 0.872 confidence and a low benign probability (0.128). The linked clinical language has more worrisome aspects (4) than reassuring features (2), resulting in a +2 net semantic influence that supports malignant classification. Text-guided attention maps exhibited stronger and more targeted activation than base attention without text guidance, highlighting irregular and asymmetric lesion sites that are clinically diagnostic of malignancy. This scenario shows how TG-CAVNet incorporates clinical semantics to magnify diagnostically essential visual cues, enabling transparent risk classification and clinically actionable decisions, such as urgent dermatological referral.

6. Conclusion

This research introduced TG-CAVNet, a text-guided cross-attentive multimodal framework for automated skin lesion diagnosis using dermoscopic images and clinical text. For clinically grounded visual reasoning, the model uses Bio-ClinicalBERT-based semantic encoding, text-guided channel-wise feature modulation, text-queried cross-attention, and adaptive multi-stream fusion. Quantitative examination of a multimodal dermoscopic dataset showed that TG-CAVNet outperformed state-of-the-art multimodal baselines with 90.75% accuracy, 87.5% F1-score, 0.8017 MCC, and 0.8016 Cohen's Kappa Clinical measures showed strong performance, with a macro Jaccard score of 0.8205 and a Hamming loss of 0.0925. Ablation experiments confirmed the contribution of each architectural component, while interpretability investigations revealed that text-guided attention increases semantic–visual alignment and transparent decision-making. In future, multiclass lesion diagnosis, patient metadata and longitudinal lesion progression may be added to TG-CAVNet. For real-time deployment, attention processes and visual backbones can be optimized for computational efficiency. Large-scale clinical validation, uncertainty estimation and clinician-in-the-loop evaluation will improve the safety and explainability of dermatological workflow integration.

Additional Information:

Data Availability: Dataset is available at the link.
https://ieee-dataport.org/documents/multimodal-dermoscopic-dataset

Conflict of interest: The authors declare no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Author contribution: All authors contributed to the study conception and design. Introduction and Related work were performed by Keerthika P and Suresh P. Problem formulation, proposed methodology and Results were done by Nitesh Kumar AR and Keerthika P. All authors read and approved the final manuscript.

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